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GitHub Link	https://github.com/120608akhila/MLA0405-DEEP-LEARNING/tree/main/Assessment/CO3

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING
SLOT : D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

*I **AKHILA POCHIMIREDDY(192425225)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AKHILA.P

Name of the student: P.AKHILA

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Answer:

A medical imaging organization wants to classify X-ray images. **CNNs are more suitable than traditional Fully Connected Neural Networks (FCNNs)** because they preserve spatial relationships and automatically learn image features.

Comparison

Parameter	Fully Connected Neural Network	CNN
Architecture	Every neuron connects to all neurons in the previous layer	Uses convolution, pooling and fully connected layers
Input handling	Image is usually flattened into a 1D vector	Preserves 2D/3D spatial structure
Feature extraction	Does not explicitly consider local image regions	Automatically extracts local features
Parameters	Very large number of weights	Fewer parameters due to parameter sharing
Spatial information	Mostly lost after flattening	Preserved through convolution layers
Computational complexity	High for high-resolution images	More efficient for images
Image suitability	Less suitable	Highly suitable
Translation handling	Poor	Better because filters detect features at different locations

Role of CNN Components

- **Convolutional layers** extract local features such as edges, lines and textures.
- **Filters** scan the image and identify specific patterns.
- **Pooling layers** reduce the size of feature maps and computational requirements.
- **Fully connected layers** use the extracted features for final classification.

Critical Evaluation

For medical X-ray images, an FCNN may require a very large number of parameters because every pixel can be connected to every neuron. CNNs use local connections and parameter sharing, reducing the number of parameters while retaining important spatial information.

Conclusion

CNNs are more suitable for X-ray classification because they **automatically extract spatial features, use fewer parameters and handle image structures efficiently.**

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Answer:

A CNN contains different layers, and each layer performs a specific task in detecting industrial defects.

Layer	Main Function	Effect
Convolution	Extracts features	Detects edges, textures and patterns
Pooling	Reduces dimensions	Decreases computation
Activation	Adds non-linearity	Learns complex patterns
Fully Connected	Classification	Predicts defect class

Convolutional Layer

The convolutional layer applies filters to different regions of an image. It detects important features such as edges, corners, textures and shapes.

Pooling Layer

Pooling reduces the spatial size of feature maps. Max pooling, for example, selects the maximum value from a small region. This reduces computation while retaining important information.

Activation Layer

Functions such as **ReLU** introduce non-linearity, allowing the network to learn complex defect patterns.

Fully Connected Layer

The extracted features are combined by the fully connected layer to classify the product as normal or defective.

Early and Deep Filters

Early-layer filters generally learn simple features such as edges, lines and textures. **Deeper-layer filters** combine these simple features and learn complex patterns such as cracks, scratches and damaged shapes.

Critical Evaluation

All layers contribute differently. Removing convolution reduces feature extraction, while excessive pooling may remove useful small-defect information. Therefore, the layers must be properly balanced.

Conclusion

The combination of **convolution, activation, pooling and classification layers** allows the CNN to progressively learn defect features and improve industrial product inspection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Answer:

Parameter sharing is an important property of CNNs. Instead of learning separate weights for every image location, CNNs reuse the same filter across different locations.

Parameter	Independent Weights	CNN Parameter Sharing
Number of parameters	Very high	Lower
Memory requirement	High	Lower
Computation	Expensive	More efficient
Feature detection	Location-dependent	Same feature at different locations
Learning efficiency	Lower	Higher
Scalability	Poor for large images	Better

Parameter Sharing in CNN

Consider a **3×3 filter** used to detect an edge. The same nine filter weights are moved across the image. Therefore, the filter can detect the same edge at different positions.

This reduces the number of trainable parameters significantly.

Advantages

1. **Reduced memory:** Fewer weights need to be stored.
2. **Lower computation:** Fewer parameters need to be learned.
3. **Efficient learning:** A useful feature is learned once and reused.
4. **Feature detection:** Objects can be detected even when their position changes.
5. **Scalability:** Suitable for large numbers of high-resolution images.

Critical Evaluation

Independent weights can learn location-specific patterns but require huge numbers of parameters. This makes them unsuitable for processing thousands of high-resolution surveillance images.

CNN parameter sharing provides better efficiency, although it does not provide complete translation invariance by itself; pooling and other CNN operations contribute to positional tolerance.

Conclusion

Parameter sharing is important because it **reduces memory, computation and training requirements while allowing the same feature detector to work across the image.**

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Answer:

AlexNet and ResNet are important CNN architectures, but ResNet is designed to solve some limitations associated with deeper networks.

Parameter	AlexNet	ResNet
Network depth	Relatively shallow	Much deeper
Organization	Sequential layers	Residual blocks
Filters	Larger early filters	Mainly smaller filters in residual blocks
Feature extraction	Good	More powerful
Training	Comparatively simpler	Designed for deep training
Gradient problem	Less applicable due to shallower depth	Skip connections improve gradient flow
Computational cost	Lower than very deep networks	Can be higher for deep versions
Complex recognition	Good	Better suited

AlexNet

AlexNet uses convolutional layers, ReLU activation, pooling and fully connected layers. It demonstrated that deep CNNs could achieve strong performance in large-scale image classification.

ResNet

ResNet introduces **skip connections** or **shortcut connections**. Instead of forcing every layer to learn a complete mapping, the network learns a residual mapping.

A simplified residual connection is:

$$\text{Output} = F(x) + x$$

The shortcut helps information and gradients pass through the network more easily.

Critical Evaluation

AlexNet has a simpler architecture and lower depth, but it is less capable of learning very complex hierarchical representations. ResNet supports much deeper networks and reduces the degradation and gradient-flow problems that occur when simply increasing network depth.

However, deeper ResNet models can require more computational resources.

Recommendation

For a complex computer-vision application, **ResNet is the better choice** because its residual connections allow deeper networks to be trained effectively and provide stronger feature extraction.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization** and a **CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Answer:

The agricultural CNN has very high training accuracy but poor performance on unseen leaf images. This indicates **overfitting**, where the model learns the training data too closely instead of learning general disease patterns.

Parameter	Non-Regularized CNN	Regularized CNN
Overfitting	High	Reduced
Generalization	Poor	Better
Training accuracy	Very high	More balanced
Test performance	Often poor	Better
Training stability	Lower	Better with suitable techniques
Real-world reliability	Lower	Higher

1. Dropout

Dropout randomly deactivates some neurons during training. It prevents the network from depending too much on particular neurons and helps reduce overfitting.

2. Data Augmentation

Data augmentation creates additional variations of existing leaf images through:

- Rotation
- Flipping
- Cropping
- Scaling
- Brightness changes

This helps the model handle different orientations and lighting conditions.

3. Batch Normalization

Batch normalization normalizes intermediate activations during training. It can improve training stability and make optimization more efficient.

Critical Evaluation

A non-regularized CNN may achieve very high training accuracy but poor validation accuracy because it memorizes the training images. Regularization improves the ability of the model to generalize to unseen leaves.

However, too much dropout or excessive image transformation can cause underfitting. Therefore, regularization should be applied carefully.

Recommended Approach

For plant disease classification, a suitable approach is:

Data Augmentation + Convolution + Batch Normalization + ReLU + Pooling + Dropout + Classification

Data augmentation provides diverse training examples, batch normalization improves training stability, and dropout reduces over-dependence on individual neurons.

Conclusion

A **regularized CNN is more reliable** for real-world plant disease classification because it reduces overfitting and improves performance on unseen images.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand